Emad Gohari

Senior Machine Learning Engineer · LLM Systems, Agentic AI, NLP

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SUMMARY

Senior machine learning engineer with **8 years of experience** taking NLP and LLM systems from research to production. At Thomson Reuters Labs, shipped **production GenAI summarization** and semantic search, contributed to a **commercially launched agentic AI product**, and built the model-retraining pipelines and **PII-secure data infrastructure** behind CLEAR's entity resolution. Turns research code into reliable, reproducible pipelines and works closely with researchers, subject matter experts, and product teams in regulated domains.

TECHNICAL SKILLS

LLM & Agentic AI: Large Language Models (LLMs), Agentic AI, Multi-Agent Orchestration, PydanticAI, Prompt Engineering, LLM Evaluation (LLM-as-Judge), GPT-4, Claude, OpenAI API

Machine Learning & NLP: Machine Learning, Deep Learning, Model Evaluation, NLP, Transformers (Hugging Face), PyTorch, scikit-learn, XGBoost, Sentence Embeddings, Semantic Search, Information Retrieval, spaCy, NLTK, pandas, NumPy

MLOps, Data & Cloud: MLOps, ML Pipelines, Experiment Tracking (MLflow), Data Pipelines, AWS SageMaker (Studio, Pipelines), Apache Spark (PySpark), Snowflake, AWS (S3, EMR), AWS CloudFormation, Docker, PostgreSQL, Elasticsearch, OpenSearch

Languages & Practices: Python, Java, Scala, Bash, Agile Development

EXPERIENCE

Senior Machine Learning Engineer II (Research) at Thomson Reuters

Jul 2021 - Present

Toronto, Canada

Applied ML at TR Labs for CLEAR, Thomson Reuters' due-diligence platform for KYC/AML screening and investigations, and the Checkpoint tax research platform. Built ML systems across the lifecycle in a regulated environment, including LLM features and agentic workflows, model-training pipelines, and secure data infrastructure.

WORK PROJECTS

- **Entity Resolution Retraining & Snowflake Pipelines: CLEAR** [INTERNAL TOOL] 2026

  Designed and built an **on-demand model-retraining pipeline** on SageMaker Pipelines, turning researchers' feature-generation, training, and evaluation code into reusable DAG steps. Enabled **parallel experiments tracked in MLflow**, cutting experiment turnaround by an **estimated ~40%**. Built proof-of-concept Snowflake processing and training pipelines and assessed the gaps for migrating AWS research workflows to Snowflake.

  Tech & Skills: Python, AWS SageMaker Pipelines, MLflow, PySpark, Snowflake, XGBoost, Poetry

- **Agentic Investigation System: CLEAR Investigate** [PRODUCTION] 2025 - 2026

  Contributed to **CLEAR Investigate, Thomson Reuters' commercially launched agentic AI product** for investigations. Built a **central configuration layer** in PydanticAI for agent system prompts, tool definitions, and agent/subagent wiring, so the team could **reconfigure investigation agents from one place** instead of editing definitions across many files. Migrated the LLM-as-judge evaluation suite into a reusable repository prepared for CI/CD.

  Tech & Skills: Python, PydanticAI, Claude, Agentic AI, LLM-as-Judge Evaluation, Prompt Engineering

- **CLEAR Business AI: GenAI Report Summarization** [PRODUCTION] 2024

  Built the GenAI summary panel for CLEAR Business entity reports, **live in production processing hundreds of reports daily**. Designed selective XML extraction feeding **targeted per-section LLM calls** (business overview, key personnel, adverse media, liens and lawsuits), plus a Bing Search + LLM verification step that **filters false matches between same-named entities**, with each claim linked to its source in the report. Refined prompts and extraction logic through SME annotation rounds against product-manager exemplars.

  Tech & Skills: Python, GPT-4, Claude Sonnet, Bing Search API, Prompt Engineering, XML Processing

- **Entity Resolution Data Infrastructure: CLEAR KYC/AML** [INTERNAL TOOL] 2022 - 2024

  Built the research data infrastructure for CLEAR's entity resolution system (~800M entities, billions of documents, 700 idents/second). Designed a **PII-isolated AWS research environment** with SageMaker Studio, CloudFormation, and per-researcher profiles; defined a **versioned schema reconciling two incompatible annotation sources**, with Spark and Python pipelines to merge records and surface label conflicts; and automated the manual workflow into **scheduled, repeatable SageMaker Pipelines**.

  Tech & Skills: Python, Apache Spark (PySpark), AWS SageMaker, AWS CloudFormation

- **Semantic Search Improvement: Checkpoint Tax Research** [PRODUCTION] 2021

  Shipped a **sentence-embedding query-intent classifier** for Checkpoint, Thomson Reuters' tax research platform, separating broad conceptual queries from specific tax lookups to promote higher-level documents. **Improved access to conceptual documents by 95%** within a **~100ms latency budget**. Identified the retrieval gap from SME and customer feedback.

  Tech & Skills: Python, Sentence Embeddings, Text Classification, Semantic Search

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NLP & Machine Learning Engineer at INAGO Inc. Jul 2018 - Jun 2021

Toronto, Canada

Built NLP and language-understanding systems end to end, covering data preparation, model training, evaluation, and application integration. Work included transformer fine-tuning and research collaborations with university and industry partners.

WORK PROJECTS

- **Automated Question Generation from Documents** [RESEARCH] 2020

  Fine-tuned **T5** to generate questions from documents, **cutting manual data-curation effort by 40%**. Experimented with input representations and evaluated generated text with BLEURT in a research collaboration with university partners.

  Tech & Skills: Python, PyTorch, Hugging Face Transformers, T5, BLEURT

- **Domain-Specific Language Understanding** [RESEARCH] 2019

  Trained domain-specific **Word2Vec embeddings** and **LSTM-based NLU models**, with interpretability testing to make model decisions transparent.

  Tech & Skills: Python, PyTorch, Word2Vec, LSTM

- **Conversational Question Answering** [RESEARCH] 2018

  Built a **conversational question-answering system** using syntactic and semantic document analysis and automatic ontology generation, in a research collaboration with an industry partner.

  Tech & Skills: Python, spaCy

EDUCATION

M.Sc. in Computer Science, York University 2014 - 2018

Toronto, Canada · GPA: 8.17 / 9

Thesis: Interactive Question Answering Using Frame-based Knowledge Representation

B.Sc. in Computer Engineering, Amirkabir University of Technology 2010 - 2014

Tehran, Iran · GPA: 17.18 / 20

PUBLICATIONS

Question-worthy Sentence Selection for Question Generation
Canadian AI 2020, 2020 · co-authored

Interactive Question Answering Using Frame-based Knowledge Representation
York University M.Sc. Thesis, 2018

Time Aware Topic-based Recommender System
Big Data & Information Analytics, 2016 · co-authored

A Study on Prediction of User's Tendency Toward Purchases in Websites based on Behavior Models
Information and Knowledge Technology (IKT), 6th Conference, IEEE, 2014 · co-authored